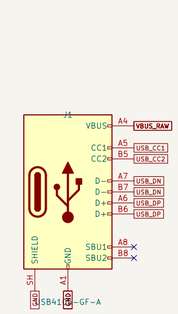
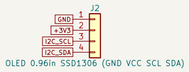
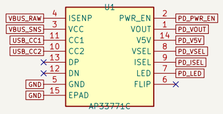
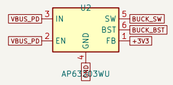
USB-C PD input (AP33771C asks for 9 V), 3.3 V buck, ESP32-C3, OLED + 3 buttons, I2C mux to the six slots.
Label-style schematic generated from design.py: pins with the same label are connected.

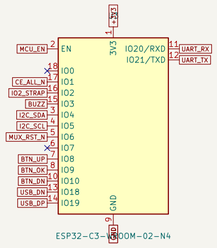






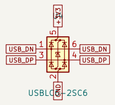


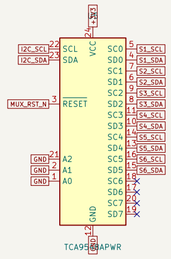


















Sheet: /lipo-charger/
File: lipo-charger.kicad_sch

Title: 6-slot 1S/2S USB-C LiPo charger — power input, MCU, UI

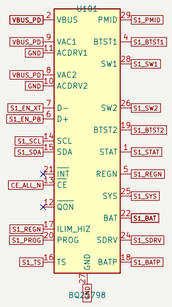
Size: A2	Date:	Rev: A
KiCad E.D.A. 10.0.8		Id: 1/14

Slot 1 · Slot 2 · Slot 3 · Slot 4 · Slot 5 · Slot 6

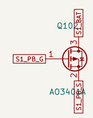
BQ25798 buck-boost charger. D+ drives the PicoBlade switch, D− the XT30 switch (register 0x47).
Each connector has a back-to-back P-FET pair so the two packs can never be connected together.

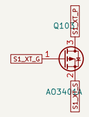


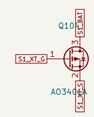










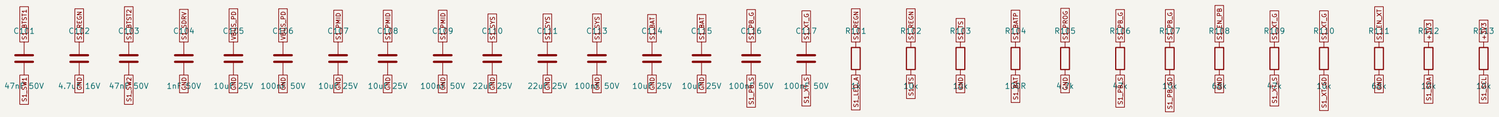








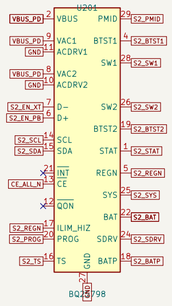




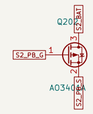
BQ25798 buck-boost charger. D+ drives the PicoBlade switch, D− the XT30 switch (register 0x47).
Each connector has a back-to-back P-FET pair so the two packs can never be connected together.

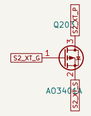


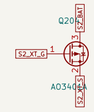




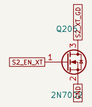






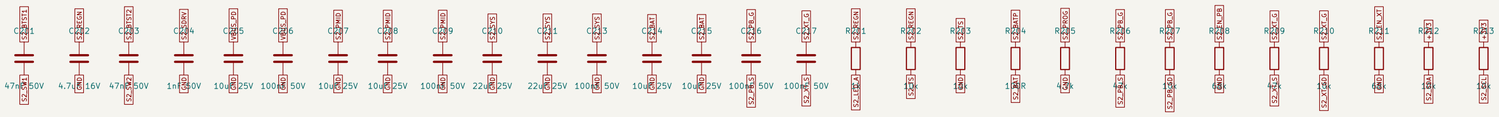








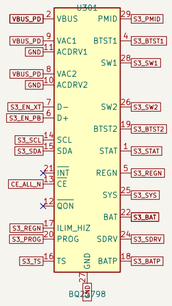




BQ25798 buck-boost charger. D+ drives the PicoBlade switch, D− the XT30 switch (register 0x47).
Each connector has a back-to-back P-FET pair so the two packs can never be connected together.

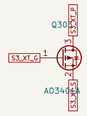


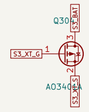










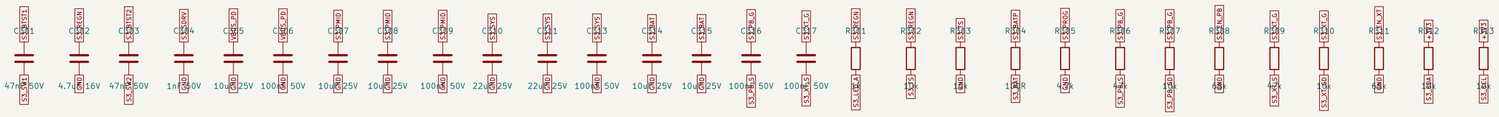












BQ25798 buck-boost charger. D+ drives the PicoBlade switch, D− the XT30 switch (register 0x47).
Each connector has a back-to-back P-FET pair so the two packs can never be connected together.

J401
XT30 2S

J402
PicoBlade 1S

U401
BQ25798

Q401
AO3401A

Q402
AO3401A

Q403
AO3401A

Q404
AO3401A

Q405
2N7002

Q406
2N7002

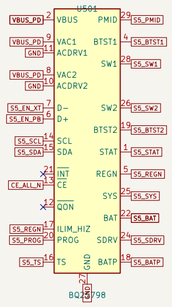
L401
2.2µH 3A

D401
CHARGING (green)

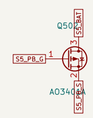
BQ25798 buck-boost charger. D+ drives the PicoBlade switch, D− the XT30 switch (register 0x47).
Each connector has a back-to-back P-FET pair so the two packs can never be connected together.

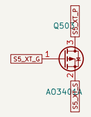


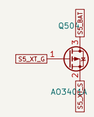










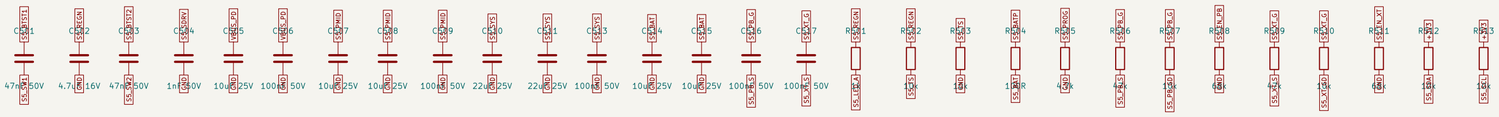












BQ25798 buck-boost charger. D+ drives the PicoBlade switch, D− the XT30 switch (register 0x47).
Each connector has a back-to-back P-FET pair so the two packs can never be connected together.

J601
XT30 2S

J602
PicoBlade 1S

U601
BQ25798

Q601
AO3401A

Q602
AO3401A

Q603
AO3401A

Q604
AO3401A

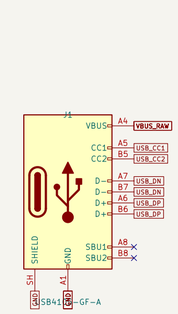
Q605
2N7002

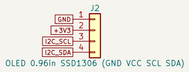
Q606
2N7002

L601
2.2µH 3A

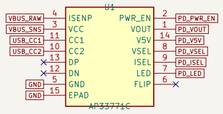
D601
CHARGING (green)

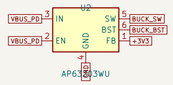
USB-C PD input (AP33771C asks for 9 V), 3.3 V buck, ESP32-C3, OLED + 3 buttons, I2C mux to the six slots.
Label-style schematic generated from design.py: pins with the same label are connected.

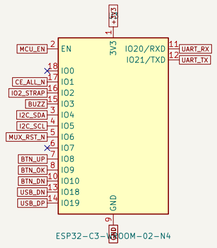






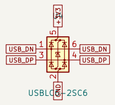


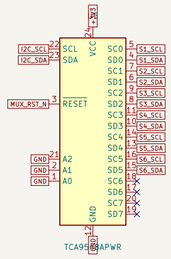










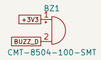






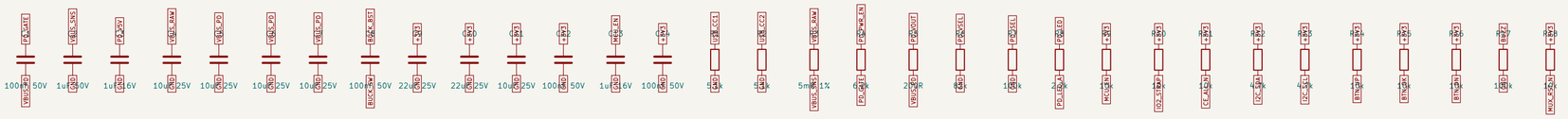








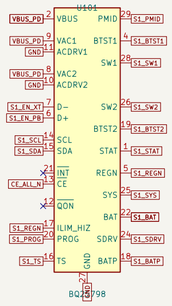



100nF 50V
1uF 50V
1uF 16V
10uF 25V
100nF 25V
10uF 25V
10uF 25V
100nF 50V
22uF 25V
22uF 25V
10uF 25V
100nF 50V
1uF 16V
100nF 50V
5mΩ 1%
2.2Ω 1%
200kΩ
100k
100k
100k
47k
10k
10k
10

BQ25798 buck-boost charger. D+ drives the PicoBlade switch, D− the XT30 switch (register 0x47).
Each connector has a back-to-back P-FET pair so the two packs can never be connected together.

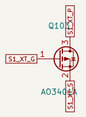


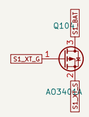










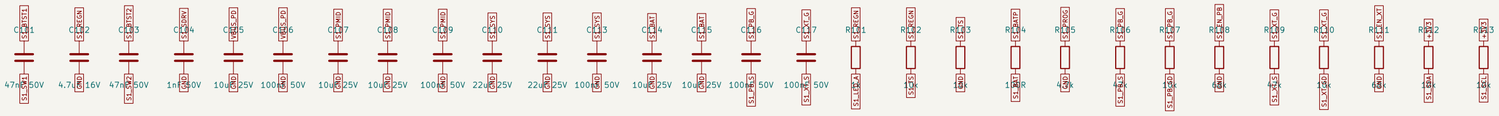












Sheet: /lipo-charger/Slot 1/
File: slot1.kicad_sch

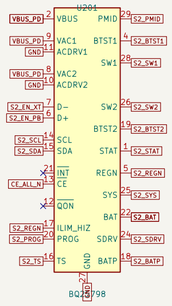
Title: Charger slot 1: BQ25798 + PicoBlade (1S) / XT30 (2S) switches

Size: A2	Date:	Rev: A
KiCad E.D.A. 10.0.8		Id: 4/14

BQ25798 buck-boost charger. D+ drives the PicoBlade switch, D− the XT30 switch (register 0x47).
Each connector has a back-to-back P-FET pair so the two packs can never be connected together.

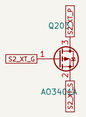


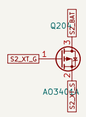










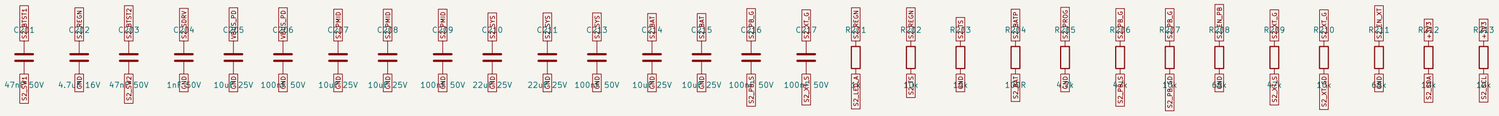












Sheet: /lipo-charger/Slot 2/
File: slot2.kicad_sch

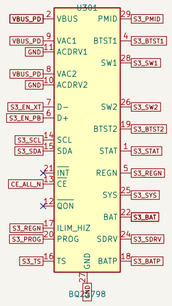
Title: Charger slot 2: BQ25798 + PicoBlade (1S) / XT30 (2S) switches

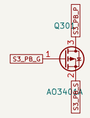
Size: A2	Date:	Rev: A
KiCad E.D.A. 10.0.8		Id: 5/14

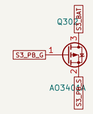
BQ25798 buck-boost charger. D+ drives the PicoBlade switch, D− the XT30 switch (register 0x47).
Each connector has a back-to-back P-FET pair so the two packs can never be connected together.

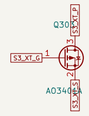


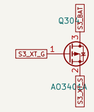










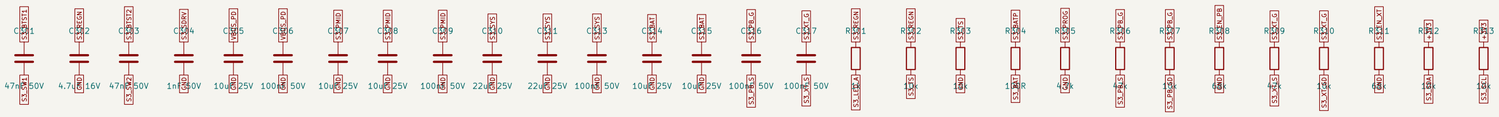












Sheet: /lipo-charger/Slot 3/
File: slot3.kicad_sch

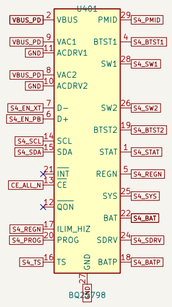
Title: Charger slot 3: BQ25798 + PicoBlade (1S) / XT30 (2S) switches

Size: A2	Date:	Rev: A
KiCad E.D.A. 10.0.8		Id: 8/14

BQ25798 buck-boost charger. D+ drives the PicoBlade switch, D− the XT30 switch (register 0x47).
Each connector has a back-to-back P-FET pair so the two packs can never be connected together.

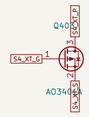


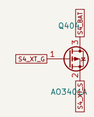










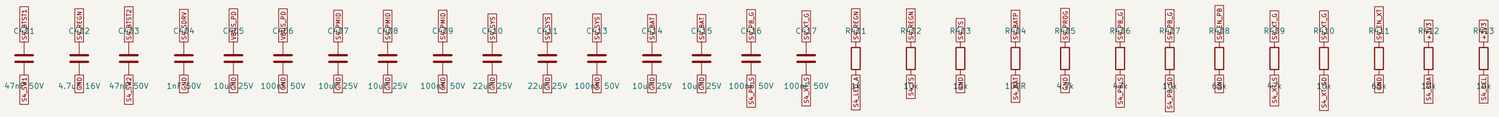








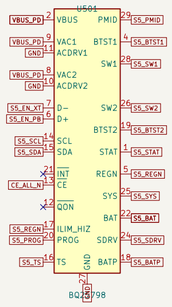




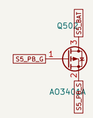
BQ25798 buck-boost charger. D+ drives the PicoBlade switch, D− the XT30 switch (register 0x47).
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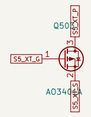


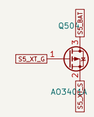










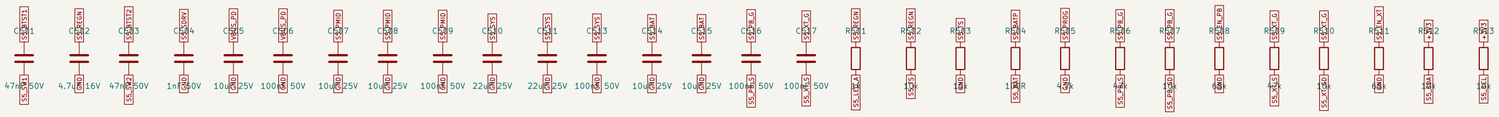












BQ25798 buck-boost charger. D+ drives the PicoBlade switch, D− the XT30 switch (register 0x47).
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J601
XT30 2S

J602
PicoBlade 1S

U601
BQ25798

Q601
AO3401A

Q602
AO3401A

Q603
AO3401A

Q604
AO3401A

Q605
2N7002

Q606
2N7002

L601
2.2µH 3A

D601
CHARGING (green)